

## Appendices

### Appendix A

In this appendix, we derive the expressions of  $Q_1$ - $Q_{10}$  and establish the inequality  $(Q_2+Q_8)^2-(Q_5+Q_{10})^2 > 0$ .

$Q_0 = 3d + 4k_1x_1^* + 4k_2x_2^* + mx_1^* + \frac{b(1-rx_2^*)x_2^*}{a+(1-rx_2^*)x_1^*} - \frac{b(1-rx_2^*)^2x_1^*x_2^*}{[a+(1-rx_2^*)x_1^*]^2}$ ,  $Q_1 = (d + k_1x_1^* + k_2x_2^* + mx_1^*)(d + 2k_1x_1^* + k_2x_2^* + \frac{b(1-rx_2^*)x_2^*}{a+(1-rx_2^*)x_1^*} - \frac{b(1-rx_2^*)^2x_1^*x_2^*}{[a+(1-rx_2^*)x_1^*]^2}) + (d + k_1x_1^* + 2k_2x_2^*)(d + 2k_1x_1^* + k_2x_2^* + \frac{b(1-rx_2^*)x_2^*}{a+(1-rx_2^*)x_1^*} - \frac{b(1-rx_2^*)^2x_1^*x_2^*}{[a+(1-rx_2^*)x_1^*]^2}) + (d + k_1x_1^* + k_2x_2^* + mx_1^*)(d + k_1x_1^* + 2k_2x_2^*) - k_1x_2^*(k_2x_1^* + \frac{(b-2brx_2^*)x_1^*}{a+(1-rx_2^*)x_1^*} + \frac{br(1-rx_2^*)x_1^{*2}x_2^*}{[a+(1-rx_2^*)x_1^*]^2})$ ,  $Q_2 = (d + k_1x_1^* + k_2x_2^* + mx_1^*)(d + 2k_1x_1^* + k_2x_2^* + \frac{b(1-rx_2^*)x_2^*}{a+(1-rx_2^*)x_1^*} - \frac{b(1-rx_2^*)^2x_1^*x_2^*}{[a+(1-rx_2^*)x_1^*]^2})(d + k_1x_1^* + 2k_2x_2^*) - k_1x_2^*(d + k_1x_1^* + k_2x_2^* + mx_1^*)(k_2x_1^* + \frac{(b-2brx_2^*)x_1^*}{a+(1-rx_2^*)x_1^*} + \frac{br(1-rx_2^*)x_1^{*2}x_2^*}{[a+(1-rx_2^*)x_1^*]^2})$ ,  $Q_3 = -ms^*$ ,  $Q_4 = -ms^*(d + k_1x_1^* + k_2x_2^* + mx_1^*) - ms^*(d + k_1x_1^* + 2k_2x_2^*) + mx_1^*(ms^* + k_1s^* - k_1s^0)$ ,  $Q_5 = -ms^*(d + k_1x_1^* + k_2x_2^* + mx_1^*)(d + k_1x_1^* + 2k_2x_2^*) + mk_1k_2(s^0 - s^*)x_1^*x_2^* + mx_1^*(ms^* + k_1s^* - k_1s^0)(d + k_1x_1^* + 2k_2x_2^*)$ ,  $Q_6 = -(\frac{(b-2brx_2^*)x_1^*}{a+(1-rx_2^*)x_1^*} + \frac{br(1-rx_2^*)x_1^{*2}x_2^*}{[a+(1-rx_2^*)x_1^*]^2})$ ,  $Q_7 = -(2d + 3k_1x_1^* + 2k_2x_2^* + mx_1^* + \frac{b(1-rx_2^*)x_2^*}{a+(1-rx_2^*)x_1^*} - \frac{b(1-rx_2^*)^2x_1^*x_2^*}{[a+(1-rx_2^*)x_1^*]^2})(\frac{(b-2brx_2^*)x_1^*}{a+(1-rx_2^*)x_1^*} + \frac{br(1-rx_2^*)x_1^{*2}x_2^*}{[a+(1-rx_2^*)x_1^*]^2}) + (k_2x_1^* + \frac{(b-2brx_2^*)x_1^*}{a+(1-rx_2^*)x_1^*} + \frac{br(1-rx_2^*)x_1^{*2}x_2^*}{[a+(1-rx_2^*)x_1^*]^2})$ ,  $Q_8 = (d + k_1x_1^* + k_2x_2^* + mx_1^*)(k_2x_1^* + \frac{(b-2brx_2^*)x_1^*}{a+(1-rx_2^*)x_1^*} + \frac{br(1-rx_2^*)x_1^{*2}x_2^*}{[a+(1-rx_2^*)x_1^*]^2}) \times (\frac{b(1-rx_2^*)x_2^*}{a+(1-rx_2^*)x_1^*} - \frac{b(1-rx_2^*)^2x_1^*x_2^*}{[a+(1-rx_2^*)x_1^*]^2}) - (d + k_1x_1^* + k_2x_2^* + mx_1^*)(d + 2k_1x_1^* + k_2x_2^* + \frac{b(1-rx_2^*)x_2^*}{a+(1-rx_2^*)x_1^*} - \frac{b(1-rx_2^*)^2x_1^*x_2^*}{[a+(1-rx_2^*)x_1^*]^2}) \times (\frac{(b-2brx_2^*)x_1^*}{a+(1-rx_2^*)x_1^*} + \frac{br(1-rx_2^*)x_1^{*2}x_2^*}{[a+(1-rx_2^*)x_1^*]^2})$ ,  $Q_9 = ms^*(\frac{(b-2brx_2^*)x_1^*}{a+(1-rx_2^*)x_1^*} + \frac{br(1-rx_2^*)x_1^{*2}x_2^*}{[a+(1-rx_2^*)x_1^*]^2})$ ,  $Q_{10} = ms^*(d + k_1x_1^* + k_2x_2^* + mx_1^*)(\frac{(b-2brx_2^*)x_1^*}{a+(1-rx_2^*)x_1^*} + \frac{br(1-rx_2^*)x_1^{*2}x_2^*}{[a+(1-rx_2^*)x_1^*]^2}) - mx_1^*k_2(s^0 - s^*)(\frac{b(1-rx_2^*)x_2^*}{a+(1-rx_2^*)x_1^*} - \frac{b(1-rx_2^*)^2x_1^*x_2^*}{[a+(1-rx_2^*)x_1^*]^2}) - mx_1^*(ms^* + k_1s^* - k_1s^0)(\frac{(b-2brx_2^*)x_1^*}{a+(1-rx_2^*)x_1^*} + \frac{br(1-rx_2^*)x_1^{*2}x_2^*}{[a+(1-rx_2^*)x_1^*]^2})$ . Based on  $Q_2 + Q_8 + Q_5 + Q_{10} > 0$  given in Theorem ??, we present a proof that  $(Q_2 + Q_8)^2 - (Q_5 + Q_{10})^2 > 0$ .

Define  $D_0 = d + k_1x_1^* + k_2x_2^* > 0$ ,  $M = mx_1^* > 0$ ,  $H = a + (1 - rx_2^*)x_1^* > 0$ ,  $X = x_1^* + x_2^* > 0$ ,  $U = \frac{b(1-2rx_2^*)x_1^*}{H} + \frac{br(1-rx_2^*)x_1^{*2}x_2^*}{H^2}$ ,  $V = \frac{b(1-rx_2^*)x_2^*}{H} - \frac{b(1-rx_2^*)^2x_1^*x_2^*}{H^2} > 0$ . From the Eq. (3.3) satisfied by the positive equilibrium, we further obtain

$$\begin{cases} D_0H = b(1 - rx_2^*)x_1^*, \\ mx_1^*s = D_0X, \\ s = s^0 - X. \end{cases}$$

Then  $Q_5 = -ms(D_0 + M)(D_0 + k_2x_2^*) + Mk_1k_2Xx_2^* + M(ms - k_1X)(D_0 + k_2x_2^*)$ ,  $Q_{10} = msD_0U - Mk_2XV + k_1MXU$ . After straightforward calculations, we obtain

$$Q_5 + Q_{10} = (U - D_0 - k_2x_2^*)(k_1MX + msD_0) + Mk_2X(k_1x_2^* - V).$$

Subsequently, we calculate

$$\begin{aligned} U - D_0 &= \frac{bx_1^*}{H} - \frac{2brx_1^*x_2^*}{H} + \frac{br(1 - rx_2^*)x_1^{*2}x_2^*}{H^2} - \frac{b(1 - rx_2^*)x_1^*}{H} \\ &= -\frac{brx_1^*x_2^*}{H} \left(1 - \frac{(1 - rx_2^*)x_1^*}{H}\right) = -\frac{abrx_1^*x_2^*}{H^2} < 0, \end{aligned}$$

and

$$\begin{aligned}
Q_5 + Q_{10} &= (U - D_0 - k_2 x_2^*)(k_1 M X + m s D_0) + M k_2 X (k_1 x_2^* - V) \\
&< -k_2 x_2^* (k_1 M X + m s D_0) + M k_2 X (k_1 x_2^* - V) \\
&< -k_2 x_2^* m s D_0 - M k_2 X V \\
&= -k_2 m s x_2^* (d + k_1 x_1^* + k_2 x_2^*) - \frac{a b m k_2 x_1^* x_2^* (x_1^* + x_2^*) (1 - r x_2^*)}{H^2} < 0.
\end{aligned}$$

Thus, we have  $Q_2 + Q_8 - (Q_5 + Q_{10}) > 0$ . Thus, we obtain  $(Q_2 + Q_8)^2 - (Q_5 + Q_{10})^2 > 0$ .

## Appendix B. The calculations of coefficients $g_{20}, g_{11}, g_{02}$ , and $g_{21}$ .

From the definition  $z(t) = \langle q^*(s), y_t \rangle$ , we obtain

$$\dot{z}(t) = i w_3 \bar{\tau}_{20} z(t) + g(z, \bar{z}), \quad (\text{B.1})$$

where

$$g(z, \bar{z}) = (\bar{q}^*(0))^T F(0, W(z, \bar{z}, \theta) + 2 \operatorname{Re}[z(t)q(\theta)]) = g_{20} \frac{z^2}{2} + g_{11} z \bar{z} + g_{02} \frac{\bar{z}^2}{2} + g_{21} \frac{z^2 \bar{z}}{2} + \dots \quad (\text{B.2})$$

From the solution  $y_t(\theta) = W(z(t), \bar{z}(t), \theta) + 2 [\operatorname{Re} z(t)q(\theta)]$  on the center manifold and the expression of  $F$ , we obtain

$$(\bar{q}^*(0))^T F(0, y_t) = D \bar{\tau}_{20} [P_7 + \alpha_2 P_8 + \beta_2 P_9], \quad (\text{B.3})$$

$$\begin{aligned}
\text{where } P_7 &= 2(p_{14}\alpha_1 - k_2\beta_1)\frac{z^2}{2} + 2(p_{14}\operatorname{Re}\{\alpha_1\} - k_2\operatorname{Re}\{\beta_1\})z\bar{z} + 2(p_{14}\bar{\alpha}_1 - k_2\bar{\beta}_1)\frac{\bar{z}^2}{2} + (2p_{14}W_{11}^{(2)}(0) + p_{14}W_{20}^{(2)}(0) \\
&+ p_{14}\bar{\alpha}_1 W_{20}^{(1)}(0) + 2p_{14}\alpha_1 W_{11}^{(1)}(0) - 2k_2 W_{11}^{(3)}(0) - k_2 W_{20}^{(3)}(0) - k_2 \bar{\beta}_1 W_{20}^{(1)}(0) - 2k_2 \beta_1 W_{11}^{(1)}(0))\frac{z^2 \bar{z}}{2}, \\
P_8 &= (2m\alpha_1 e^{-2i\omega_3 \bar{\tau}_1} + 2p_{25}\alpha_1^2 + 2p_{26}\alpha_1\beta_1 + 2p_{27}\beta_1^2)\frac{z^2}{2} \\
&+ (m\bar{\alpha}_1 e^{i\omega_3 \bar{\tau}_1} + m\alpha_1 + 2p_{25}\alpha_1\bar{\alpha}_1 + 2p_{26}\operatorname{Re}\{\alpha_1\bar{\beta}_1\} + 2p_{27}\beta_1\bar{\beta}_1)z\bar{z} + (2m\bar{\alpha}_1 e^{2i\omega_3 \bar{\tau}_1} + 2p_{25}\bar{\alpha}_1^2 + 2p_{26}\bar{\alpha}_1\bar{\beta}_1 + 2p_{27}\bar{\beta}_1^2)\frac{\bar{z}^2}{2} \\
&+ (2m e^{-i\omega_3 \bar{\tau}_1} W_{11}^{(2)}(-\frac{\bar{\tau}_1}{\tau_{20}}) + m e^{i\omega_3 \bar{\tau}_1} W_{20}^{(2)}(-\frac{\bar{\tau}_1}{\tau_{20}}) + m\bar{\alpha}_1 e^{i\omega_3 \bar{\tau}_1} W_{20}^{(1)}(-\frac{\bar{\tau}_1}{\tau_{20}}) + 2m\alpha_1 W_{11}^{(1)}(-\frac{\bar{\tau}_1}{\tau_{20}}) + 2p_{25}\alpha_1 W_{11}^{(2)}(0) \\
&+ 2p_{25}\bar{\alpha}_1 W_{20}^{(2)}(0) + 2p_{25}\alpha_1 W_{11}^{(3)}(0) + 2p_{26}\alpha_1 W_{11}^{(3)}(0) + p_{26}\bar{\alpha}_1 W_{20}^{(3)}(0) + p_{26}\bar{\beta}_1 W_{20}^{(2)}(0) + 2p_{26}\beta_1 W_{11}^{(2)}(0) + 2p_{27}\beta_1 W_{11}^{(3)}(0) \\
&+ 2p_{27}\bar{\beta}_1 W_{20}^{(3)}(0) + 2p_{27}\beta_1 W_{11}^{(3)}(0))\frac{z^2 \bar{z}}{2}, \\
P_9 &= (2p_{35}\alpha_1\beta_1 e^{-2i\omega_3 \tau_{20}} + 2p_{36}\alpha_1^2 e^{-2i\omega_3 \tau_{20}} - 2k_1\alpha_1\beta_1 - 2k_2\beta_1^2 + 2p_{37}\beta_1^2 e^{-2i\omega_3 \tau_{20}})\frac{z^2}{2} + (2p_{35}\operatorname{Re}\{\alpha_1\bar{\beta}_1\} + 2p_{36}\alpha_1\bar{\alpha}_1 - \\
&2k_1\operatorname{Re}\{\alpha_1\bar{\beta}_1\} - 2k_2\beta_1\bar{\beta}_1 + 2p_{37}\beta_1\bar{\beta}_1)z\bar{z} + (2p_{35}\bar{\alpha}_1\bar{\beta}_1 e^{2i\omega_3 \tau_{20}} + 2p_{36}\bar{\alpha}_1^2 e^{2i\omega_3 \tau_{20}} - 2k_1\bar{\alpha}_1\bar{\beta}_1 - 2k_2\bar{\beta}_1^2 + 2p_{37}\bar{\beta}_1^2 e^{2i\omega_3 \tau_{20}})\frac{\bar{z}^2}{2} \\
&+ (2p_{35}\alpha_1 e^{-i\omega_3 \tau_{20}} W_{11}^{(3)}(-1) + p_{35}\bar{\alpha}_1 e^{i\omega_3 \tau_{20}} W_{20}^{(3)}(-1) + p_{35}\bar{\beta}_1 e^{i\omega_3 \tau_{20}} W_{20}^{(2)}(-1) + 2p_{35}\beta_1 e^{-i\omega_3 \tau_{20}} W_{11}^{(2)}(-1) + \\
&4p_{36}\alpha_1 e^{-i\omega_3 \tau_{20}} W_{11}^{(2)}(-1) + 2p_{36}\bar{\alpha}_1 e^{i\omega_3 \tau_{20}} W_{20}^{(2)}(-1) - 2k_1\alpha_1 W_{11}^{(3)}(0) - k_1\bar{\alpha}_1 W_{20}^{(3)}(0) - k_1\bar{\beta}_1 W_{20}^{(2)}(0) - k_1\beta_1 W_{11}^{(2)}(0) - \\
&4k_2\beta_1 W_{11}^{(3)}(0) - 2k_2\bar{\beta}_1 W_{20}^{(3)}(0) + 4p_{37}\beta_1 e^{-i\omega_3 \tau_{20}} W_{11}^{(3)}(-1) + 2p_{37}\bar{\beta}_1 e^{i\omega_3 \tau_{20}} W_{20}^{(3)}(-1))\frac{z^2 \bar{z}}{2}.
\end{aligned}$$

From Eqs. (B.2) and (B.3), we derive

$$\begin{aligned}
g_{20} &= 2D\bar{\tau}_{20}[p_{14}\alpha_1 - k_2\beta_1 + \alpha_2(m\alpha_1 e^{-2i\omega_3 \bar{\tau}_1} + p_{25}\alpha_1^2 + p_{26}\alpha_1\beta_1 + p_{27}\beta_1^2) + \beta_2(p_{35}\alpha_1\beta_1 e^{-2i\omega_3 \tau_{20}} + p_{36}\alpha_1^2 e^{-2i\omega_3 \tau_{20}} \\
&- k_1\alpha_1\beta_1 - k_2\beta_1^2 + p_{37}\beta_1^2 e^{-2i\omega_3 \tau_{20}})], \\
g_{11} &= 2D\bar{\tau}_{20}[p_{14}\operatorname{Re}\{\alpha_1\} - k_2\operatorname{Re}\{\beta_1\} + \alpha_2(\frac{1}{2}m\bar{\alpha}_1 e^{i\omega_3 \bar{\tau}_1} + \frac{1}{2}m\alpha_1 + p_{25}\alpha_1\bar{\alpha}_1 + p_{26}\operatorname{Re}\{\alpha_1\bar{\beta}_1\} + p_{27}\beta_1\bar{\beta}_1) + \\
&\beta_2(p_{35}\operatorname{Re}\{\alpha_1\bar{\beta}_1\} + p_{36}\alpha_1\bar{\alpha}_1 - k_1\operatorname{Re}\{\alpha_1\bar{\beta}_1\} - k_2\beta_1\bar{\beta}_1 + p_{37}\beta_1\bar{\beta}_1)], \\
g_{02} &= 2D\bar{\tau}_{20}[p_{14}\bar{\alpha}_1 - k_2\bar{\beta}_1 + \alpha_2(m\bar{\alpha}_1 e^{2i\omega_3 \bar{\tau}_1} + p_{25}\bar{\alpha}_1^2 + p_{26}\bar{\alpha}_1\bar{\beta}_1 + p_{27}\bar{\beta}_1^2) + \beta_2(p_{35}\bar{\alpha}_1\bar{\beta}_1 e^{2i\omega_3 \tau_{20}} + p_{36}\bar{\alpha}_1^2 e^{2i\omega_3 \tau_{20}} \\
&- k_1\bar{\alpha}_1\bar{\beta}_1 - k_2\bar{\beta}_1^2 + p_{37}\bar{\beta}_1^2 e^{2i\omega_3 \tau_{20}})], \text{ and} \\
g_{21} &= D\bar{\tau}_{20}[2p_{14}W_{11}^{(2)}(0) + p_{14}W_{20}^{(2)}(0) + p_{14}\bar{\alpha}_1 W_{20}^{(1)}(0) + 2p_{14}\alpha_1 W_{11}^{(1)}(0) - 2k_2 W_{11}^{(3)}(0) - k_2 W_{20}^{(3)}(0) - k_2 \bar{\beta}_1 W_{20}^{(1)}(0) - \\
&2k_2 \beta_1 W_{11}^{(1)}(0) + \alpha_2[2m e^{-i\omega_3 \bar{\tau}_1} W_{11}^{(2)}(-\frac{\bar{\tau}_1}{\tau_{20}}) + m e^{i\omega_3 \bar{\tau}_1} W_{20}^{(2)}(-\frac{\bar{\tau}_1}{\tau_{20}}) + m\bar{\alpha}_1 e^{i\omega_3 \bar{\tau}_1} W_{20}^{(1)}(-\frac{\bar{\tau}_1}{\tau_{20}}) + 2m\alpha_1 W_{11}^{(1)}(-\frac{\bar{\tau}_1}{\tau_{20}}) + \\
&2p_{25}\alpha_1 W_{11}^{(2)}(0) + 2p_{25}\bar{\alpha}_1 W_{20}^{(2)}(0) + 2p_{25}\alpha_1 W_{11}^{(3)}(0) + 2p_{26}\alpha_1 W_{11}^{(3)}(0) + p_{26}\bar{\alpha}_1 W_{20}^{(3)}(0) + p_{26}\bar{\beta}_1 W_{20}^{(2)}(0) + \\
&2p_{26}\beta_1 W_{11}^{(2)}(0) + 2p_{27}\beta_1 W_{11}^{(3)}(0) + 2p_{27}\bar{\beta}_1 W_{20}^{(3)}(0) + 2p_{27}\beta_1 W_{11}^{(3)}(0)] + \beta_2[2p_{35}\alpha_1 e^{-i\omega_3 \tau_{20}} W_{11}^{(3)}(-1) + \\
&p_{35}\bar{\alpha}_1 e^{i\omega_3 \tau_{20}} W_{20}^{(3)}(-1) + p_{35}\bar{\beta}_1 e^{i\omega_3 \tau_{20}} W_{20}^{(2)}(-1) + 2p_{35}\beta_1 e^{-i\omega_3 \tau_{20}} W_{11}^{(2)}(-1) + 4p_{36}\alpha_1 e^{-i\omega_3 \tau_{20}} W_{11}^{(2)}(-1) +
\end{aligned}$$

$$2p_{36}\bar{\alpha}_1 e^{i\omega_3\bar{\tau}_{20}}W_{20}^{(2)}(-1) - 2k_1\alpha_1 W_{11}^{(3)}(0) - k_1\bar{\alpha}_1 W_{20}^{(3)}(0) - k_1\bar{\beta}_1 W_{20}^{(2)}(0) - k_1\beta_1 W_{11}^{(2)}(0) - 4k_2\beta_1 W_{11}^{(3)}(0) - 2k_2\bar{\beta}_1 W_{20}^{(3)}(0) + 4p_{37}\beta_1 e^{-i\omega_3\bar{\tau}_{20}}W_{11}^{(3)}(-1) + 2p_{37}\bar{\beta}_1 e^{i\omega_3\bar{\tau}_{20}}W_{20}^{(3)}(-1)]\}.$$

As  $g_{21}$  given in the above expression contains  $W_{20}(\theta)$  and  $W_{11}(\theta)$ , we proceed to calculate these two functions. By the definition of  $W(z, \bar{z}, \theta)$ , we obtain

$$\dot{W} = \dot{y}_t - \dot{z}q(\theta) - \dot{\bar{z}}\bar{q}(\theta) = \mathcal{A}(0)W + H(z, \bar{z}, \theta), \quad (\text{B.4})$$

where

$$H(z, \bar{z}, \theta) = \begin{cases} -2 \operatorname{Re} \left\{ (\bar{q}^*(0))^T F(0, z, \bar{z}) q(\theta) \right\}, & \theta \in [-1, 0), \\ -2 \operatorname{Re} \left\{ (\bar{q}^*(0))^T F(0, z, \bar{z}) q(\theta) \right\} + F(0, z, \bar{z}), & \theta = 0. \end{cases} \quad (\text{B.5})$$

We now represent  $H(z, \bar{z}, \theta)$  by the series expansion below

$$H(z, \bar{z}, \theta) = H_{20}(\theta) \frac{z^2}{2} + H_{11}(\theta) z \bar{z} + H_{02}(\theta) \frac{\bar{z}^2}{2} + \dots$$

In addition, the Eqs. (B.1) and (B.2) yield

$$\dot{W} = i\omega_3 \bar{\tau}_{20} W_{20}(\theta) z^2 - i\omega_3 \bar{\tau}_{20} W_{02}(\theta) \bar{z}^2 + \dots \quad (\text{B.6})$$

Thus, the Eqs. (B.4) and (B.6) imply that

$$\begin{cases} (\mathcal{A}(0) - 2i\omega_3 \bar{\tau}_{20}) W_{20}(\theta) = -H_{20}(\theta), \\ \mathcal{A}(0) W_{11}(\theta) = -H_{11}(\theta). \end{cases} \quad (\text{B.7})$$

In what follows, we solve Eq. (B.7) to find  $W_{20}$  and  $W_{11}$ . For  $\theta \in [-1, 0)$ , it holds that

$$H(z, \bar{z}, \theta) = -(g_{20}q(\theta) + \bar{g}_{02}\bar{q}(\theta)) \frac{z^2}{2} - (g_{11}q(\theta) + \bar{g}_{11}\bar{q}(\theta)) z \bar{z} - (g_{02}q(\theta) + \bar{g}_{02}\bar{q}(\theta)) \frac{\bar{z}^2}{2} + \dots$$

Combining the definition of operator  $\mathcal{A}(0)$  with the above identity, we obtain

$$\begin{cases} \dot{W}_{20}(\theta) = 2i\omega_3 \bar{\tau}_{20} W_{20}(\theta) + g_{20}q(0)e^{i\omega_3 \bar{\tau}_{20}\theta} + \bar{g}_{02}\bar{q}(0)e^{-i\omega_3 \bar{\tau}_{20}\theta}, \\ \dot{W}_{11}(\theta) = g_{11}q(0)e^{i\omega_3 \bar{\tau}_{20}\theta} + \bar{g}_{11}\bar{q}(0)e^{-i\omega_3 \bar{\tau}_{20}\theta}. \end{cases} \quad (\text{B.8})$$

By solving Eq. (B.8), we arrive at

$$\begin{cases} W_{20}(\theta) = \frac{ig_{20}}{\omega_3 \bar{\tau}_{20}} q(0)e^{i\omega_3 \bar{\tau}_{20}\theta} + \frac{i\bar{g}_{02}}{3\omega_3 \bar{\tau}_{20}} \bar{q}(0)e^{-i\omega_3 \bar{\tau}_{20}\theta} + C_5 e^{2i\omega_3 \bar{\tau}_{20}\theta}, \\ W_{11}(\theta) = -\frac{ig_{11}}{\omega_3 \bar{\tau}_{20}} q(0)e^{i\omega_3 \bar{\tau}_{20}\theta} + \frac{i\bar{g}_{11}}{w_3 \bar{\tau}_{20}} \bar{q}(0)e^{-i\omega_3 \bar{\tau}_{20}\theta} + C_6, \end{cases} \quad (\text{B.9})$$

where  $C_5$  and  $C_6$  are undetermined constant vectors, which can be determined by setting  $\theta = 0$  and using the condition of Eq. (B.5) at  $\theta = 0$ . In this case, we have

$$H(z, \bar{z}, 0) = -(g_{20}q(0) + \bar{g}_{02}\bar{q}(0)) \frac{z^2}{2} - (g_{11}q(0) + \bar{g}_{11}\bar{q}(0)) z \bar{z} - (g_{02}q(0) + \bar{g}_{02}\bar{q}(0)) \frac{\bar{z}^2}{2} + F(0, z, \bar{z}) + \dots \quad (\text{B.10})$$

From Eq. (B.10) and the function  $F(0, z, \bar{z})$ , we obtain

$$\begin{cases} H_{20}(0) = -g_{20}q(0) - \bar{g}_{02}\bar{q}(0) + \bar{\tau}_{20}C_7, \\ H_{11}(0) = -g_{11}q(0) - \bar{g}_{11}\bar{q}(0) + \bar{\tau}_{20}C_8, \end{cases} \quad (\text{B.11})$$

where

$$C_7 = \begin{pmatrix} 2(p_{14}\alpha_1 - k_2\beta_1) \\ 2(m\alpha_1 e^{-2i\omega_3\bar{\tau}_1} + p_{25}\alpha_1^2 + p_{26}\alpha_1\beta_1 + p_{27}\beta_1^2) \\ 2(p_{35}\alpha_1\beta_1 e^{-2i\omega_3\bar{\tau}_{20}} + p_{36}\alpha_1^2 e^{-2i\omega_3\bar{\tau}_{20}} - k_1\alpha_1\beta_1 - k_2\beta_1^2 + p_{37}\beta_1^2 e^{-2i\omega_3\bar{\tau}_{20}}) \end{pmatrix},$$

and

$$C_8 = \begin{pmatrix} 2(p_{14}\operatorname{Re}\{\alpha_1\} - k_2\operatorname{Re}\{\beta_1\}) \\ m\bar{\alpha}_1 e^{i\omega_3\bar{\tau}_1} + m\alpha_1 + 2p_{25}\alpha_1\bar{\alpha}_1 + 2p_{26}\operatorname{Re}\{\alpha_1\bar{\beta}_1\} + 2p_{27}\beta_1\bar{\beta}_1 \\ 2(p_{35}\operatorname{Re}\{\alpha_1\bar{\beta}_1\} + p_{36}\alpha_1\bar{\alpha}_1 - k_1\operatorname{Re}\{\alpha_1\bar{\beta}_1\} - k_2\beta_1\bar{\beta}_1 + p_{37}\beta_1\bar{\beta}_1) \end{pmatrix}.$$

According to the definition of  $\mathcal{A}(0)$  evaluated at  $\theta = 0$

$$\begin{cases} \mathcal{A}(0)W_{20}(0) = \int_{-1}^0 d\varrho(s,0)W_{20}(s) = 2i\omega_3\bar{\tau}_{20}W_{20}(0) + g_{20}q(0) + \bar{g}_{02}\bar{q}(0) - \bar{\tau}_{20}C_7, \\ \mathcal{A}(0)W_{11}(0) = \int_{-1}^0 d\varrho(s,0)W_{11}(s) = g_{11}q(0) + \bar{g}_{11}\bar{q}(0) - \bar{\tau}_{20}C_8. \end{cases} \quad (\text{B.12})$$

By virtue of Eq. (B.9), we solve Eq. (B.12) and get that

$$C_5 = \begin{pmatrix} 2i\omega_3 - p_{11} & -p_{12} & -p_{13} \\ -p_{21}e^{-2i\omega_3\bar{\tau}_1} & 2i\omega_3 - p_{23} - p_{22}e^{-2i\omega_3\bar{\tau}_1} & -p_{24} \\ 0 & -p_{32} - p_{31}e^{-2i\omega_3\bar{\tau}_{20}} & 2i\omega_3 - p_{34} - p_{33}e^{-2i\omega_3\bar{\tau}_{20}} \end{pmatrix}^{-1} C_7,$$

and

$$C_6 = - \begin{pmatrix} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} + p_{23} & p_{24} \\ 0 & p_{32} + p_{31} & p_{33} + p_{34} \end{pmatrix}^{-1} C_8.$$

With the coefficients  $C_5$  and  $C_6$  derived above, we are able to determine the key quantities  $g_{20}$ ,  $g_{11}$ ,  $g_{02}$ ,  $g_{21}$ .

## Appendix C

In this appendix, we derive explicit expressions for  $\dot{p}_{11}$ ,  $\dot{p}_{12}$ ,  $\dot{p}_{13}$ ,  $\dot{p}_{21}$ ,  $\dot{p}_{22}$ ,  $\dot{p}_{23}$ ,  $\dot{p}_{24}$ ,  $\dot{p}_{31} + \dot{p}_{32}$ ,  $\dot{p}_{33} + \dot{p}_{34}$ .

Since the explicit closed-form solution of the positive equilibrium is unavailable, we adopt implicit differentiation together with the chain rule for subsequent calculations. Let  $\dot{x}_1 = \frac{dx_1}{d\mu_1}|_{\mu_1=0}$  and  $\dot{x}_2 = \frac{dx_2}{d\mu_1}|_{\mu_1=0}$ . From Eq. (3.3), it follows that

$$\begin{aligned} \dot{x}_1 &= \frac{\bar{x}_1^*\bar{x}_2^*(d - b + k_1\bar{x}_1^* + k_2\bar{x}_2^*)(d + (k_1 + k_2 + m)\bar{x}_1^* + 2k_2\bar{x}_2^*)}{E}, \\ \dot{x}_2 &= -\frac{\bar{x}_1^*\bar{x}_2^*(d - b + k_1\bar{x}_1^* + k_2\bar{x}_2^*)(d - ms^0 + 2(k_1 + m)\bar{x}_1^* + (k_1 + k_2 + m)\bar{x}_2^*)}{E}, \end{aligned}$$

where

$$\begin{aligned} E &= C_0 + C_{1x}\bar{x}_1^* + C_{1y}\bar{x}_2^* + C_{2x}(\bar{x}_1^*)^2 + C_{2y}(\bar{x}_2^*)^2 + C_{2xy}\bar{x}_1^*\bar{x}_2^* \\ &\quad + C_{3x}(\bar{x}_1^*)^3 + C_{3y}(\bar{x}_2^*)^3 + C_{3x^2y}(\bar{x}_1^*)^2\bar{x}_2^* + C_{3xy^2}\bar{x}_1^*(\bar{x}_2^*)^2, \end{aligned}$$

with coefficients  $C_0 = d^2 - bd + ad(k_1 - k_2) + ak_2ms^0$ ,  $C_{1x} = ak_1^2 + d^2r_0 - b(k_1 + k_2 + m) + 3dk_1 + dm - ak_1k_2 - bdr_0 + ak_1m - 2ak_2m + k_2ms^0 + bmr_0s^0 - dmr_0s^0$ ,  $C_{1y} = -ak_2^2 - d^2r_0 - 2bk_2 + 3dk_2 + ak_1k_2 + bdr_0 - ak_2m$ ,  $C_{2x} = 2k_1^2 + 2k_1m - 2k_2m - 2bk_1r_0 - 2bmr_0 + 3dk_1r_0 + 2dmr_0 - k_1mr_0s^0$ ,  $C_{2y} = 2k_2^2 + 2bk_2r_0 - 3dk_2r_0$ ,  $C_{2xy} = 4k_1k_2 - 2dk_1r_0 + 2dk_2r_0 - 2k_2mr_0s^0$ ,  $C_{3x} = 2k_1r_0(k_1 + m)$ ,  $C_{3y} = -2k_2^2r_0$ ,  $C_{3x^2y} = r_0(-k_1^2 + 3k_1k_2 - k_1m + 4k_2m)$ ,  $C_{3xy^2} = r_0(k_2^2 - 3k_1k_2 + k_2m)$ . Then, we have  $\dot{p}_{11} = -(k_1 + m)\dot{x}_1 - k_2\dot{x}_2$ ,  $\dot{p}_{12} = (k_1 +$

$$\begin{aligned}
& m)(\dot{x}_1 + \dot{x}_2), \dot{p}_{13} = k_2(\dot{x}_1 + \dot{x}_2), \dot{p}_{21} = m\dot{x}_1, \dot{p}_{22} = -m(\dot{x}_1 + \dot{x}_2), \dot{p}_{23} = -\left\{ 2k_1\dot{x}_1 + k_2\dot{x}_2 + \frac{ab\dot{x}_2}{[a+(1-r_0\bar{x}_2^*)\bar{x}_1^*]^2} - \right. \\
& \left. \frac{ab[(\dot{x}_1 + ar_0\dot{x}_2)2\bar{x}_2^* + (a-\bar{x}_1^* + r_0\bar{x}_1^*\bar{x}_2^* + 2r_0^2\bar{x}_2^*\dot{x}_1 - 4r_0\dot{x}_1)(\bar{x}_2^*)^2]}{[a+(1-r_0\bar{x}_2^*)\bar{x}_1^*]^3} \right\}, \dot{p}_{24} = -k_2\dot{x}_1 + \frac{ab\bar{x}_1^*(2ar_0\dot{x}_2 - \dot{x}_1 + 2r_0\bar{x}_1^*\dot{x}_2)}{[a+(1-r_0\bar{x}_2^*)\bar{x}_1^*]^3} + \\
& \frac{ab[\bar{x}_1^*\bar{x}_2^*(2\bar{x}_1^* + 2a + r_0\dot{x}_1) - a(1-2r_0\bar{x}_2^*)\dot{x}_1]}{[a+(1-r_0\bar{x}_2^*)\bar{x}_1^*]^3}, \dot{p}_{31} + \dot{p}_{32} = \frac{ab\dot{x}_2}{[a+(1-r_0\bar{x}_2^*)\bar{x}_1^*]^2} - \frac{ab[(\dot{x}_1 + ar_0\dot{x}_2)2\bar{x}_2^* + (a-\bar{x}_1^* + r_0\bar{x}_1^*\bar{x}_2^* + 2r_0^2\bar{x}_2^*\dot{x}_1 - 4r_0\dot{x}_1)(\bar{x}_2^*)^2]}{[a+(1-r_0\bar{x}_2^*)\bar{x}_1^*]^3} \\
& - k_1\dot{x}_2, \dot{p}_{33} + \dot{p}_{34} = -\frac{ab\bar{x}_1^*(2ar_0\dot{x}_2 - \dot{x}_1 + 2r_0\bar{x}_1^*\dot{x}_2)}{[a+(1-r_0\bar{x}_2^*)\bar{x}_1^*]^3} - \frac{ab[\bar{x}_1^*\bar{x}_2^*(2\bar{x}_1^* + 2a + r_0\dot{x}_1) - a(1-2r_0\bar{x}_2^*)\dot{x}_1]}{[a+(1-r_0\bar{x}_2^*)\bar{x}_1^*]^3} - (k_1\dot{x}_1 + 2k_2\dot{x}_2), \text{ where } \bar{x}_1^* \\
& \text{and } \bar{x}_2^* \text{ denote the values of } x_1^* \text{ and } x_2^* \text{ evaluated at } r = r_0.
\end{aligned}$$

## Appendix D

The coefficients  $D_{11}, D_{12}, D_{31}$ , and  $D_{32}$  are computed as:

$$\begin{aligned}
D_{11} &= \frac{1}{i\omega_1} \left( -2\psi_1^T(0)A_{2000}\psi_1^T(0)A_{1100} + \psi_1^T(0)A_{1100}\psi_1^T(0)A_{2000} + \psi_1^T(0)A_{1100}\psi_2^T(0)A_{1100} + \frac{2}{3}\psi_1^T(0)A_{0200}\psi_2^T(0)A_{2000} - \right. \\
& \left. \frac{m}{n}\psi_1^T(0)A_{1010}\psi_3^T(0)A_{1100} - \frac{m}{n-2m}\psi_1^T(0)A_{0110}\psi_3^T(0)A_{2000} + \frac{m}{n}\psi_1^T(0)A_{1001}\psi_4^T(0)A_{1100} + \frac{m}{n+2m}\psi_1^T(0)A_{0101}\psi_3^T(0)A_{2000} \right), \\
D_{12} &= \frac{1}{i\omega_1} \left( -2\psi_1^T(0)A_{2000}\psi_1^T(0)A_{0011} - \frac{m}{n}\psi_1^T(0)A_{1010}\psi_1^T(0)A_{1001} + \frac{m}{n}\psi_1^T(0)A_{1001}\psi_1^T(0)A_{1001} + \psi_1^T(0)A_{1100}\psi_2^T(0)A_{0011} - \right. \\
& \left. \frac{m}{n-2m}\psi_1^T(0)A_{0110}\psi_2^T(0)A_{1001} + \frac{m}{n+2m}\psi_1^T(0)A_{0101}\psi_2^T(0)A_{1010} - \frac{m}{n}\psi_1^T(0)A_{1010}\psi_3^T(0)A_{0011} - \frac{2m}{2n-m}\psi_1^T(0)A_{0020}\psi_3^T(0)A_{1001} + \right. \\
& \left. \psi_1^T(0)A_{0011}\psi_3^T(0)A_{1010} + \frac{m}{n}\psi_1^T(0)A_{1001}\psi_4^T(0)A_{0011} + \psi_1^T(0)A_{0011}\psi_4^T(0)A_{1001} + \frac{2m}{2n+m}\psi_1^T(0)A_{0002}\psi_4^T(0)A_{1010} \right), \\
D_{31} &= \frac{1}{i\omega_1} \left( -\psi_3^T(0)A_{1010}\psi_1^T(0)A_{0011} + \frac{m}{2n-m}\psi_3^T(0)A_{1001}\psi_1^T(0)A_{0020} + \psi_3^T(0)A_{0110}\psi_2^T(0)A_{0011} + \frac{m}{2n+m}\psi_3^T(0)A_{0101} \times \right. \\
& \left. \psi_2^T(0)A_{0020} - \frac{2m}{n}\psi_3^T(0)A_{0020}\psi_3^T(0)A_{0011} + \frac{m}{n}\psi_3^T(0)A_{0011}\psi_3^T(0)A_{0020} + \frac{m}{n}\psi_3^T(0)A_{0011}\psi_4^T(0)A_{0011} + \frac{2m}{3n}\psi_3^T(0)A_{0002} \times, \right. \\
& \left. \psi_4^T(0)A_{0020} \right), \\
D_{32} &= \frac{1}{i\omega_1} \left( \frac{2m}{n-2m}\psi_3^T(0)A_{2000}\psi_1^T(0)A_{0011} + \frac{m}{n}\psi_3^T(0)A_{1100}\psi_1^T(0)A_{1010} - \psi_3^T(0)A_{1010}\psi_1^T(0)A_{1100} + \frac{m}{n}\psi_3^T(0)A_{1100}\psi_2^T(0)A_{0110} \right. \\
& \left. + \frac{2m}{n+2m}\psi_3^T(0)A_{0200}\psi_2^T(0)A_{1010} + \psi_3^T(0)A_{0110}\psi_2^T(0)A_{1100} - \psi_3^T(0)A_{1010}\psi_3^T(0)A_{0110} + \psi_3^T(0)A_{0110}\psi_3^T(0)A_{1010} - \right. \\
& \left. \frac{2m}{n}\psi_3^T(0)A_{0020}\psi_3^T(0)A_{1100} + \frac{m}{2n-m}\psi_3^T(0)A_{1001}\psi_4^T(0)A_{0110} + \frac{m}{2n+m}\psi_3^T(0)A_{0101}\psi_4^T(0)A_{1010} + \frac{m}{n}\psi_3^T(0)A_{0011}\psi_4^T(0)A_{1100} \right),
\end{aligned}$$

$$\text{where } A_{2000} = \tau_0 \begin{pmatrix} p_{14}\alpha_3 - k_2\beta_3 \\ m\alpha_3 e^{-2i\omega_1\tau_0} + \bar{p}_{25}\alpha_3^2 + \bar{p}_{26}\alpha_3\beta_3 + \bar{p}_{27}\beta_3^2 \\ (\bar{p}_{35} - k_1)\alpha_3\beta_3 + \bar{p}_{36}\alpha_3^2 + (\bar{p}_{37} - k_2)\beta_3^2 \end{pmatrix}, \quad A_{0200} = \overline{A_{2000}},$$

$$A_{0020} = \tau_0 \begin{pmatrix} p_{14}\alpha_4 - k_2\beta_4 \\ m\alpha_4 e^{-2i\omega_2\tau_0} + \bar{p}_{25}\alpha_4^2 + \bar{p}_{26}\alpha_4\beta_4 + \bar{p}_{27}\beta_4^2 \\ (\bar{p}_{35} - k_1)\alpha_4\beta_4 + \bar{p}_{36}\alpha_4^2 + (\bar{p}_{37} - k_2)\beta_4^2 \end{pmatrix}, \quad A_{0002} = \overline{A_{0020}},$$

$$A_{1100} = 2\tau_0 \begin{pmatrix} p_{14} \operatorname{Re}\{\bar{\alpha}_3\} - k_2 \operatorname{Re}\{\bar{\beta}_3\} \\ m \operatorname{Re}\{\bar{\alpha}_3\} + \bar{p}_{25} \operatorname{Re}\{\alpha_3\bar{\alpha}_3\} + \bar{p}_{26} \operatorname{Re}\{\alpha_3\bar{\beta}_3\} + \bar{p}_{27} \operatorname{Re}\{\beta_3\bar{\beta}_3\} \\ (\bar{p}_{35} - k_1) \operatorname{Re}\{\alpha_3\bar{\beta}_3\} + \bar{p}_{36} \operatorname{Re}\{\alpha_3\bar{\alpha}_3\} + (\bar{p}_{37} - k_2) \operatorname{Re}\{\beta_3\bar{\beta}_3\} \end{pmatrix},$$

$$A_{0011} = 2\tau_0 \begin{pmatrix} p_{14} \operatorname{Re}\{\bar{\alpha}_4\} - k_2 \operatorname{Re}\{\bar{\beta}_4\} \\ m \operatorname{Re}\{\bar{\alpha}_4\} + \bar{p}_{25} \operatorname{Re}\{\alpha_4\bar{\alpha}_4\} + \bar{p}_{26} \operatorname{Re}\{\alpha_4\bar{\beta}_4\} + \bar{p}_{27} \operatorname{Re}\{\beta_4\bar{\beta}_4\} \\ (\bar{p}_{35} - k_1) \operatorname{Re}\{\alpha_4\bar{\beta}_4\} + \bar{p}_{36} \operatorname{Re}\{\alpha_4\bar{\alpha}_4\} + (\bar{p}_{37} - k_2) \operatorname{Re}\{\beta_4\bar{\beta}_4\} \end{pmatrix},$$

$$A_{1010} = \tau_0 \begin{pmatrix} p_{14}(\alpha_4 + \alpha_3) - k_2(\beta_4 + \beta_3) \\ m(\alpha_4 e^{-i(\omega_1 + \omega_2)\tau_0} + \alpha_3 e^{-i(\omega_1 + \omega_2)\tau_0}) + 2\bar{p}_{25}\alpha_3\alpha_4 + \bar{p}_{26}(\alpha_3\beta_4 + \alpha_4\beta_3) + 2\bar{p}_{27}\beta_3\beta_4 \\ (\bar{p}_{35} - k_1)(\alpha_3\beta_4 + \alpha_4\beta_3) + 2\bar{p}_{36}\alpha_3\alpha_4 + 2(\bar{p}_{37} - k_2)\beta_3\beta_4 \end{pmatrix}, \quad A_{0101} = \overline{A_{1010}},$$

$$A_{0110} = \tau_0 \begin{pmatrix} p_{14}(\alpha_4 + \bar{\alpha}_3) - k_2(\beta_4 + \bar{\beta}_3) \\ m(\alpha_4 e^{i(\omega_1 - \omega_2)\tau_0} + \bar{\alpha}_3 e^{i(\omega_1 - \omega_2)\tau_0}) + 2\bar{p}_{25}\bar{\alpha}_3\alpha_4 + \bar{p}_{26}(\bar{\alpha}_3\beta_4 + \alpha_4\bar{\beta}_3) + 2\bar{p}_{27}\bar{\beta}_3\beta_4 \\ (\bar{p}_{35} - k_1)(\bar{\alpha}_3\beta_4 + \alpha_4\bar{\beta}_3) + 2\bar{p}_{36}\bar{\alpha}_3\alpha_4 + 2(\bar{p}_{37} - k_2)\bar{\beta}_3\beta_4 \end{pmatrix}, \quad A_{1001} = \overline{A_{0110}}.$$

Moreover, we have

$$\begin{aligned}
A_{2100} &= \tau_0 \begin{pmatrix} 0 \\ 3\bar{p}_{38}\alpha_3^2\bar{\alpha}_3 + \bar{p}_{39}(\alpha_3^2\bar{\beta}_3 + 2\alpha_3\bar{\alpha}_3\beta_3) + \bar{p}_{40}(\bar{\alpha}_3\beta_3^2 + 2\alpha_3\beta_3\bar{\beta}_3) + 3\bar{p}_{41}\beta_3^2\bar{\beta}_3 \\ -3\bar{p}_{38}\alpha_3^2\bar{\alpha}_3 - \bar{p}_{39}(\alpha_3^2\bar{\beta}_3 + 2\alpha_3\bar{\alpha}_3\beta_3) - \bar{p}_{40}(\bar{\alpha}_3\beta_3^2 + 2\alpha_3\beta_3\bar{\beta}_3) - 3\bar{p}_{41}\beta_3^2\bar{\beta}_3 \end{pmatrix}, \\
A_{1011} &= 2\tau_0 \begin{pmatrix} 0 \\ 3\bar{p}_{38}\alpha_3\alpha_4\bar{\alpha}_4 + \bar{p}_{39}(\alpha_3\alpha_4\bar{\beta}_4 + \alpha_3\bar{\alpha}_4\beta_4 + \alpha_4\bar{\alpha}_4\beta_4) + \bar{p}_{40}(\alpha_3\beta_4\bar{\beta}_4 + \alpha_4\beta_3\bar{\beta}_4 + \beta_3\beta_4\bar{\alpha}_4) + 3\bar{p}_{41}\beta_3\beta_4\bar{\beta}_4 \\ -3\bar{p}_{38}\alpha_3\alpha_4\bar{\alpha}_4 - \bar{p}_{39}(\alpha_3\alpha_4\bar{\beta}_4 + \alpha_3\bar{\alpha}_4\beta_4 + \alpha_4\bar{\alpha}_4\beta_4) - \bar{p}_{40}(\alpha_3\beta_4\bar{\beta}_4 + \alpha_4\beta_3\bar{\beta}_4 + \beta_3\beta_4\bar{\alpha}_4) - 3\bar{p}_{41}\beta_3\beta_4\bar{\beta}_4 \end{pmatrix}, \\
A_{0021} &= \tau_0 \begin{pmatrix} 0 \\ 3\bar{p}_{38}\alpha_4^2\bar{\alpha}_4 + \bar{p}_{39}(\alpha_4^2\bar{\beta}_4 + 2\alpha_4\bar{\alpha}_4\beta_4) + \bar{p}_{40}(\bar{\alpha}_4\beta_4^2 + 2\alpha_4\beta_4\bar{\beta}_4) + 3\bar{p}_{41}\beta_4^2\bar{\beta}_4 \\ -3\bar{p}_{38}\alpha_4^2\bar{\alpha}_4 - \bar{p}_{39}(\alpha_4^2\bar{\beta}_4 + 2\alpha_4\bar{\alpha}_4\beta_4) - \bar{p}_{40}(\bar{\alpha}_4\beta_4^2 + 2\alpha_4\beta_4\bar{\beta}_4) - 3\bar{p}_{41}\beta_4^2\bar{\beta}_4 \end{pmatrix}, \\
A_{1110} &= 2\tau_0 \begin{pmatrix} 0 \\ 3\bar{p}_{38}\alpha_3\alpha_4\bar{\alpha}_3 + \bar{p}_{39}(\alpha_3\alpha_4\bar{\beta}_3 + \alpha_4\bar{\alpha}_3\beta_3 + \alpha_3\bar{\alpha}_3\beta_3) + \bar{p}_{40}(\alpha_4\beta_3\bar{\beta}_3 + \alpha_3\beta_4\bar{\beta}_3 + \beta_4\beta_3\bar{\alpha}_3) + 3\bar{p}_{41}\beta_3\beta_4\bar{\beta}_3 \\ -3\bar{p}_{38}\alpha_3\alpha_4\bar{\alpha}_3 - \bar{p}_{39}(\alpha_3\alpha_4\bar{\beta}_3 + \alpha_4\bar{\alpha}_3\beta_3 + \alpha_3\bar{\alpha}_3\beta_3) - \bar{p}_{40}(\alpha_4\beta_3\bar{\beta}_3 + \alpha_3\beta_4\bar{\beta}_3 + \beta_4\beta_3\bar{\alpha}_3) - 3\bar{p}_{41}\beta_3\beta_4\bar{\beta}_3 \end{pmatrix}.
\end{aligned}$$

## Appendix E

In this appendix, we present the detailed derivations of  $h_{2000}$ ,  $h_{0020}$ ,  $h_{1100}$ ,  $h_{0011}$ ,  $h_{1010}$  and  $h_{0110}$ . From (??), we conclude that  $h_{2000}(\theta)$ ,  $h_{0020}(\theta)$ ,  $h_{1100}(\theta)$ ,  $h_{0011}(\theta)$ ,  $h_{1010}(\theta)$  and  $h_{0110}(\theta)$  respectively satisfy the following initial-value differential equations:

$$\begin{aligned}
&\begin{cases} \dot{h}_{2000} - 2i\omega_1\tau_0 h_{2000} = \Phi\Psi(0)A_{2000}, \\ \dot{h}_{2000}(0) - L_0(h_{2000}) = A_{2000}, \end{cases} & \begin{cases} \dot{h}_{0020} - 2i\omega_2\tau_0 h_{0020} = \Phi\Psi(0)A_{0020}, \\ \dot{h}_{0020}(0) - L_0(h_{0020}) = A_{0020}, \end{cases} \\
&\begin{cases} \dot{h}_{1100} = \Phi\Psi(0), \\ \dot{h}_{1100}(0) - L_0(h_{1100}) = A_{1100}, \end{cases} & \begin{cases} \dot{h}_{0011} = \Phi\Psi(0)A_{0011}, \\ \dot{h}_{0011}(0) - L_0(h_{0011}) = A_{0011}, \end{cases} \\
&\begin{cases} \dot{h}_{1010} - \left(\frac{n}{m} + 1\right)i\omega_1\tau_0 h_{1010} = \Phi\Psi(0)A_{1010}, \\ \dot{h}_{1010}(0) - L_0(h_{1010}) = A_{1010}, \end{cases} & \begin{cases} \dot{h}_{0110} - \left(\frac{n}{m} - 1\right)i\omega_1\tau_0 h_{0110} = \Phi\Psi(0)A_{0110}, \\ \dot{h}_{0110}(0) - L_0(h_{0110}) = A_{0110}. \end{cases}
\end{aligned}$$

By solving these six equations, we obtain

$$h_{2000}(\theta) = e^{2i\omega_1\tau_0\theta} \int_0^\theta e^{-2i\omega_1\tau_0 t} \Phi(t)\Psi(0)A_{2000} dt + C_1 e^{2i\omega_1\tau_0\theta},$$

where

$$C_1 = (2i\omega_1\tau_0 I - L(e^{2i\omega_1\tau_0\theta}))^{-1} \left[ (I - \Phi(0)\Psi(0))A_{2000} - \tau_0 N e^{-2i\omega_1\tau_0} \int_{-1}^0 e^{-2i\omega_1\tau_0 t} \Phi(t) dt \Psi(0)A_{2000} \right].$$

$$h_{0020}(\theta) = e^{2i\omega_2\tau_0\theta} \int_0^\theta e^{-2i\omega_2\tau_0 t} \Phi(t)\Psi(0)A_{0020} dt + C_2 e^{2i\omega_2\tau_0\theta},$$

where

$$C_2 = (2i\omega_2\tau_0 I - L(e^{2i\omega_2\tau_0\theta}))^{-1} \left[ (I - \Phi(0)\Psi(0))A_{0020} - \tau_0 N e^{-2i\omega_2\tau_0} \int_{-1}^0 e^{-2i\omega_2\tau_0 t} \Phi(t) dt \Psi(0)A_{0020} \right].$$

$$h_{1100}(\theta) = \int_0^\theta \Phi(t)\Psi(0)A_{1100} dt + C_3,$$

where

$$C_3 = -(\tau_0(M + N))^{-1} \left[ (I - \Phi(0)\Psi(0))A_{1100} - \tau_0 N \int_{-1}^0 \Phi(t) dt \Psi(0)A_{1100} \right].$$

$$h_{0011}(\theta) = \int_0^\theta \Phi(t) \Psi(0) A_{0011} dt + C_4,$$

where

$$C_4 = -(\tau_0(M + N))^{-1} \left[ (I - \Phi(0) \Psi(0)) A_{0011} - \tau_0 N \int_{-1}^0 \Phi(t) dt \Psi(0) A_{0011} \right].$$

$$h_{1010}(\theta) = e^{i(\frac{n}{m}+1)\omega_1\tau_0\theta} \int_0^\theta e^{-i(\frac{n}{m}+1)\omega_1\tau_0t} \Phi(t) \Psi(0) A_{1010} dt + C_5 e^{i(\frac{n}{m}+1)\omega_1\tau_0\theta},$$

where

$$C_5 = \left( i \left( \frac{n}{m} + 1 \right) \omega_1 \tau_0 I - L \left( e^{i(\frac{n}{m}+1)\omega_1\tau_0\theta} \right) \right)^{-1} \\ \times \left[ (I - \Phi(0) \Psi(0)) A_{1010} - \tau_0 N e^{-i(\frac{n}{m}+1)\omega_1\tau_0} \int_{-1}^0 e^{-i(\frac{n}{m}+1)\omega_1\tau_0t} \Phi(t) dt \Psi(0) A_{1010} \right].$$

$$h_{0110}(\theta) = e^{i(\frac{n}{m}-1)\omega_1\tau_0\theta} \int_0^\theta e^{-i(\frac{n}{m}-1)\omega_1\tau_0t} \Phi(t) \Psi(0) A_{0110} dt + C_6 e^{i(\frac{n}{m}-1)\omega_1\tau_0\theta},$$

where

$$C_6 = \left( i \left( \frac{n}{m} - 1 \right) \omega_1 \tau_0 I - L \left( e^{i(\frac{n}{m}-1)\omega_1\tau_0\theta} \right) \right)^{-1} \\ \times \left[ (I - \Phi(0) \Psi(0)) A_{0110} - \tau_0 N e^{-i(\frac{n}{m}-1)\omega_1\tau_0} \int_{-1}^0 e^{-i(\frac{n}{m}-1)\omega_1\tau_0t} \Phi(t) dt \Psi(0) A_{0110} \right].$$

Furthermore, a direct calculation yields

$$H_2(\Phi x, v) = \tau_0 \begin{pmatrix} p_{14} (\varphi_1(0) v^{(2)}(0) + \varphi_2(0) v^{(1)}(0)) - k_2 (\varphi_1(0) v^{(3)}(0) + \varphi_3(0) v^{(1)}(0)) \\ m (\varphi_1(-1) v^{(2)}(-1) + \varphi_2(-1) v^{(1)}(-1)) + 2\bar{p}_{25} \varphi_2(0) v^{(2)}(0) \\ + \bar{p}_{26} (\varphi_2(0) v^{(3)}(0) + \varphi_3(0) v^{(2)}(0)) + 2\bar{p}_{27} \varphi_3(0) v^{(3)}(0) \\ (\bar{p}_{35} - k_1) (\varphi_2(0) v^{(3)}(0) + \varphi_3(0) v^{(2)}(0)) + 2\bar{p}_{36} \varphi_2(0) v^{(2)}(0) \\ + 2(\bar{p}_{37} - k_2) \varphi_3(0) v^{(3)}(0) \end{pmatrix}.$$

Then, from it, one obtains

$$H_2(\phi_1, h_{1100}) \\ = \tau_0 \begin{pmatrix} (p_{14}\alpha_3 - k_2\beta_3)h_{1100}^{(1)}(0) + p_{14}h_{1100}^{(2)}(0) - k_2h_{1100}^{(3)}(0) \\ m\alpha_3 e^{-i\omega_1\tau_0} h_{1100}^{(1)}(-1) + (2\bar{p}_{25}\alpha_3 + \bar{p}_{26}\beta_3)h_{1100}^{(2)}(0) + m e^{-i\omega_1\tau_0} h_{1100}^{(2)}(-1) + (\bar{p}_{26}\alpha_3 + 2\bar{p}_{27}\beta_3)h_{1100}^{(3)}(0) \\ ((\bar{p}_{35} - k_1)\beta_3 + 2\bar{p}_{36}\alpha_3)h_{1100}^{(2)}(0) + ((\bar{p}_{35} - k_1)\alpha_3 + 2(\bar{p}_{37} - k_2)\beta_3)h_{1100}^{(3)}(0) \end{pmatrix},$$

$$H_2(\phi_2, h_{2000}) \\ = \tau_0 \begin{pmatrix} (p_{14}\bar{\alpha}_3 - k_2\bar{\beta}_3)h_{2000}^{(1)}(0) + p_{14}h_{2000}^{(2)}(0) - k_2h_{2000}^{(3)}(0) \\ m\bar{\alpha}_3 e^{i\omega_1\tau_0} h_{2000}^{(1)}(-1) + (2\bar{p}_{25}\bar{\alpha}_3 + \bar{p}_{26}\bar{\beta}_3)h_{2000}^{(2)}(0) + m e^{i\omega_1\tau_0} h_{2000}^{(2)}(-1) + (\bar{p}_{26}\bar{\alpha}_3 + 2\bar{p}_{27}\bar{\beta}_3)h_{2000}^{(3)}(0) \\ ((\bar{p}_{35} - k_1)\bar{\beta}_3 + 2\bar{p}_{36}\bar{\alpha}_3)h_{2000}^{(2)}(0) + ((\bar{p}_{35} - k_1)\bar{\alpha}_3 + 2(\bar{p}_{37} - k_2)\bar{\beta}_3)h_{2000}^{(3)}(0) \end{pmatrix},$$

$$H_2(\phi_1, h_{0011}) \\ = \tau_0 \begin{pmatrix} (p_{14}\alpha_3 - k_2\beta_3)h_{0011}^{(1)}(0) + p_{14}h_{0011}^{(2)}(0) - k_2h_{0011}^{(3)}(0) \\ m\alpha_3 e^{-i\omega_1\tau_0} h_{0011}^{(1)}(-1) + (2\bar{p}_{25}\alpha_3 + \bar{p}_{26}\beta_3)h_{0011}^{(2)}(0) + m e^{-i\omega_1\tau_0} h_{0011}^{(2)}(-1) + (\bar{p}_{26}\alpha_3 + 2\bar{p}_{27}\beta_3)h_{0011}^{(3)}(0) \\ ((\bar{p}_{35} - k_1)\beta_3 + 2\bar{p}_{36}\alpha_3)h_{0011}^{(2)}(0) + ((\bar{p}_{35} - k_1)\alpha_3 + 2(\bar{p}_{37} - k_2)\beta_3)h_{0011}^{(3)}(0) \end{pmatrix},$$

$$H_2(\phi_3, h_{1001})$$

$$\begin{aligned}
&= \tau_0 \begin{pmatrix} (p_{14}\alpha_4 - k_2\beta_4)h_{1001}^{(1)}(0) + p_{14}h_{1001}^{(2)}(0) - k_2h_{1001}^{(3)}(0) \\ m\alpha_4 e^{-i\omega_2\tau_0} h_{1001}^{(1)}(-1) + (2\bar{p}_{25}\alpha_4 + \bar{p}_{26}\beta_4)h_{1001}^{(2)}(0) + me^{-i\omega_2\tau_0} h_{1001}^{(2)}(-1) + (\bar{p}_{26}\alpha_4 + 2\bar{p}_{27}\beta_4)h_{1001}^{(3)}(0) \\ ((\bar{p}_{35} - k_1)\beta_4 + 2\bar{p}_{36}\alpha_4)h_{1001}^{(2)}(0) + ((\bar{p}_{35} - k_1)\alpha_4 + 2(\bar{p}_{37} - k_2)\beta_4)h_{1001}^{(3)}(0) \end{pmatrix}, \\
H_2(\phi_4, h_{1010}) \\
&= \tau_0 \begin{pmatrix} (p_{14}\bar{\alpha}_4 - k_2\bar{\beta}_4)h_{1010}^{(1)}(0) + p_{14}h_{1010}^{(2)}(0) - k_2h_{1010}^{(3)}(0) \\ m\bar{\alpha}_4 e^{i\omega_2\tau_0} h_{1010}^{(1)}(-1) + (2\bar{p}_{25}\bar{\alpha}_4 + \bar{p}_{26}\bar{\beta}_4)h_{1010}^{(2)}(0) + me^{i\omega_2\tau_0} h_{1010}^{(2)}(-1) + (\bar{p}_{26}\bar{\alpha}_4 + 2\bar{p}_{27}\bar{\beta}_4)h_{1010}^{(3)}(0) \\ ((\bar{p}_{35} - k_1)\bar{\beta}_4 + 2\bar{p}_{36}\bar{\alpha}_4)h_{1010}^{(2)}(0) + ((\bar{p}_{35} - k_1)\bar{\alpha}_4 + 2(\bar{p}_{37} - k_2)\bar{\beta}_4)h_{1010}^{(3)}(0) \end{pmatrix}, \\
H_2(\phi_3, h_{0011}) \\
&= \tau_0 \begin{pmatrix} (p_{14}\alpha_4 - k_2\beta_4)h_{0011}^{(1)}(0) + p_{14}h_{0011}^{(2)}(0) - k_2h_{0011}^{(3)}(0) \\ m\alpha_4 e^{-i\omega_2\tau_0} h_{0011}^{(1)}(-1) + (2\bar{p}_{25}\alpha_4 + \bar{p}_{26}\beta_4)h_{0011}^{(2)}(0) + me^{-i\omega_2\tau_0} h_{0011}^{(2)}(-1) + (\bar{p}_{26}\alpha_4 + 2\bar{p}_{27}\beta_4)h_{0011}^{(3)}(0) \\ ((\bar{p}_{35} - k_1)\beta_4 + 2\bar{p}_{36}\alpha_4)h_{0011}^{(2)}(0) + ((\bar{p}_{35} - k_1)\alpha_4 + 2(\bar{p}_{37} - k_2)\beta_4)h_{0011}^{(3)}(0) \end{pmatrix}, \\
H_2(\phi_4, h_{0020}) \\
&= \tau_0 \begin{pmatrix} (p_{14}\bar{\alpha}_4 - k_2\bar{\beta}_4)h_{0020}^{(1)}(0) + p_{14}h_{0020}^{(2)}(0) - k_2h_{0020}^{(3)}(0) \\ m\bar{\alpha}_4 e^{i\omega_2\tau_0} h_{0020}^{(1)}(-1) + (2\bar{p}_{25}\bar{\alpha}_4 + \bar{p}_{26}\bar{\beta}_4)h_{0020}^{(2)}(0) + me^{i\omega_2\tau_0} h_{0020}^{(2)}(-1) + (\bar{p}_{26}\bar{\alpha}_4 + 2\bar{p}_{27}\bar{\beta}_4)h_{0020}^{(3)}(0) \\ ((\bar{p}_{35} - k_1)\bar{\beta}_4 + 2\bar{p}_{36}\bar{\alpha}_4)h_{0020}^{(2)}(0) + ((\bar{p}_{35} - k_1)\bar{\alpha}_4 + 2(\bar{p}_{37} - k_2)\bar{\beta}_4)h_{0020}^{(3)}(0) \end{pmatrix}, \\
H_2(\phi_1, h_{0110}) \\
&= \tau_0 \begin{pmatrix} (p_{14}\alpha_3 - k_2\beta_3)h_{0110}^{(1)}(0) + p_{14}h_{0110}^{(2)}(0) - k_2h_{0110}^{(3)}(0) \\ m\alpha_3 e^{-i\omega_1\tau_0} h_{0110}^{(1)}(-1) + (2\bar{p}_{25}\alpha_3 + \bar{p}_{26}\beta_3)h_{0110}^{(2)}(0) + me^{-i\omega_1\tau_0} h_{0110}^{(2)}(-1) + (\bar{p}_{26}\alpha_3 + 2\bar{p}_{27}\beta_3)h_{0110}^{(3)}(0) \\ ((\bar{p}_{35} - k_1)\beta_3 + 2\bar{p}_{36}\alpha_3)h_{0110}^{(2)}(0) + ((\bar{p}_{35} - k_1)\alpha_3 + 2(\bar{p}_{37} - k_2)\beta_3)h_{0110}^{(3)}(0) \end{pmatrix}, \\
H_2(\phi_2, h_{1010}) \\
&= \tau_0 \begin{pmatrix} (p_{14}\bar{\alpha}_3 - k_2\bar{\beta}_3)h_{1010}^{(1)}(0) + p_{14}h_{1010}^{(2)}(0) - k_2h_{1010}^{(3)}(0) \\ m\bar{\alpha}_3 e^{i\omega_1\tau_0} h_{1010}^{(1)}(-1) + (2\bar{p}_{25}\bar{\alpha}_3 + \bar{p}_{26}\bar{\beta}_3)h_{1010}^{(2)}(0) + me^{i\omega_1\tau_0} h_{1010}^{(2)}(-1) + (\bar{p}_{26}\bar{\alpha}_3 + 2\bar{p}_{27}\bar{\beta}_3)h_{1010}^{(3)}(0) \\ ((\bar{p}_{35} - k_1)\bar{\beta}_3 + 2\bar{p}_{36}\bar{\alpha}_3)h_{1010}^{(2)}(0) + ((\bar{p}_{35} - k_1)\bar{\alpha}_3 + 2(\bar{p}_{37} - k_2)\bar{\beta}_3)h_{1010}^{(3)}(0) \end{pmatrix}, \\
H_2(\phi_3, h_{1100}) \\
&= \tau_0 \begin{pmatrix} (p_{14}\alpha_4 - k_2\beta_4)h_{1100}^{(1)}(0) + p_{14}h_{1100}^{(2)}(0) - k_2h_{1100}^{(3)}(0) \\ m\alpha_4 e^{-i\omega_2\tau_0} h_{1100}^{(1)}(-1) + (2\bar{p}_{25}\alpha_4 + \bar{p}_{26}\beta_4)h_{1100}^{(2)}(0) + me^{-i\omega_2\tau_0} h_{1100}^{(2)}(-1) + (\bar{p}_{26}\alpha_4 + 2\bar{p}_{27}\beta_4)h_{1100}^{(3)}(0) \\ ((\bar{p}_{35} - k_1)\beta_4 + 2\bar{p}_{36}\alpha_4)h_{1100}^{(2)}(0) + ((\bar{p}_{35} - k_1)\alpha_4 + 2(\bar{p}_{37} - k_2)\beta_4)h_{1100}^{(3)}(0) \end{pmatrix}, \\
\text{where } h_{0200} = \overline{h_{2000}}, h_{0020} = \overline{h_{0002}}, h_{1100} = \overline{h_{1100}}, h_{0011} = \overline{h_{0011}}, h_{0101} = \overline{h_{1010}} \text{ and } h_{1001} = \overline{h_{0110}}.
\end{aligned}$$